Elliott Foster

Education

The University of Texas at Dallas
B.S., Computer Science - May 2007

Experience

Optum

Eden Prairie, MN (Remote)
Director, Software Engineering August 2024 - present

Accomplishments

- Championed adoption of AIDLC practices for engineering teams transforming to AI-first delivery teams
- Coordinated sunset of legacy Risk View application and progressive migration to Optum Risk Analytics replacement, resulting in $2.5M incremental savings and $7M annual ongoing savings
- Championed cloud-to-cloud migration from legacy Episource AWS to Optum AWS instance with zero customer downtime
- Led integration of Episource engineering teams into Optum
- Worked with departmental leaders to evolve Episource practices and procedures to align with Optum/UHG
- Co-led competitive analysis and proof of concept for enterprise data governance
- Oversaw data quality enforcement using Great Expectations to enhance real-time data pipeline reliability, aligning with HIPAA and internal data governance standards

Responsibilities

- Cross-functional leader for over 120 engineers across multiple teams and products
- Manager of teams through Team Leads and Engineering Managers
- Driver of departmental process changes
- Product backlog development and delivery on key milestones

Episource

Gardena, CA (Remote)
Director of Engineering August 2022 - August 2024

Accomplishments

- Implemented engineering Key Performance Indicators (KPIs) based on DevOps Research and Assessment (DORA) metrics, driving a culture of continuous improvement and accountability within the engineering department
- Spearheaded the implementation of DAST testing as part of CI/CD pipelines, significantly enhancing the security posture of the company's software products while minimizing release cycle times
- Championed the adoption of feature flagging techniques utilizing LaunchDarkly, enabling seamless feature rollouts, A/B testing, and targeted feature releases, resulting in improved product iteration cycles
- Partnered with other engineering leaders to complete HITRUST certification for the organization
- Grew first level managers in effective leadership skills and team growth strategies
- Delivered first-to-market solution for risk adjustment using the CMS-HCC Model V28

Responsibilities

- Cross-functional leader for multiple engineering teams
- Manager of teams through Team Leads and Engineering Managers
- Driver of departmental process changes
- Product backlog development and delivery on key milestones

Software Engineering Manager August 2021 - August 2022

Accomplishments

- Grew individual contributors to be better engineers
- Delivered key feature enhancements for the Episource Analyst product using React, GraphQL, and Nest.js

Responsibilities

- Work with product to prioritize development efforts
- Manage and mentor direct reports
- Work with peer managers to improve department and company-wide processes
- Lead team in development of coding, security, and deployment standards for cloud-native applications

Lead Software Engineer March 2020 - August 2021

Accomplishments

- Led the development and deployment of the Episource Analyst for Medicaid product using React, GraphQL, and Nest.js

Responsibilities

- Manager and team lead for both Medicare and Medicaid epiAnalyst teams
- Leading adoption of GraphQL for Episource APIs

Four Kitchens

Austin, TX
Architect December 2010 - March 2020

Accomplishments

- Team lead to internal developers, responsible for weekly 1 on 1's, unblocking teammates, and aiding in goal setting and professional development.
- Technical lead for the NBC project, overseeing architectural, technological, and best practices decisions.
- Presented about the NBC team's use of Amazon Neptune at **AWS re:Invent 2019**
- Progressively migrated NBC's primary data store from a document store to a graph database.
- Scaled NBC's API traffic from a few thousand requests per day to tens of millions of requests per day.
- Architected backend for frontend (BFF) micro APIs for encapsulating complicated business logic server side and allowing for lighter front ends.
- Designed and built node.js powered data pipeline to ingest content from multiple CMSs and surface data through a REST API that powers all NBC applications as well as **NBC.com**.
- Implemented the node.js based REST API that powers the Saturday Night Live **iOS app** and **website**.
- Architected and built REST API for the **Emmy Award** winning **The Tonight Show with Jimmy Fallon** and NBC.com websites.
- Developed and administered **Integrating Node.js and Drupal** training for DrupalCon Portland 2013.
- Built **Drupal Poetry**, a responsive magnetic poetry web app.
- Redeployed the search system for The Economist using hosted Apache Solr.

Responsibilities

- Managing and mentoring a team of engineers on professional growth and performance.
- Design, build, and maintain custom systems for clients.
- Diagnose and fix performance problems on large scale distributed systems.
- Train clients in using and building high performance distributed systems.

New Media Broadcasting Company/ Direct2Care

Glendale, CA

Head of Web Development July 2005 - December 2010

Accomplishments

- Deployed Varnish reverse proxy and static file cache for MashCast Drupal system.
- Deployed Apache Solr faceted search for MashCast Drupal system.
- Deployed memcached and MySQL master-slave servers for scalability of the MashCast Drupal system.
- Wrote data synchronization code to keep information from a multi-site network and a desktop client server in sync
- Wrote code to manage, re-format, and display uploaded files such as images, videos, and audio
- Created a profile based search module
- Wrote a horoscopes module that populates its current astrological sign information via RSS feeds
- Worked with “themers” (HTML/CSS experts) to ensure the output from Drupal modules was easy to handle at the theme layer

Responsibilities

- Write Drupal modules using PHP with MySQL
- Manage the release process from automated daily builds to release candidate to production release
- Create monitoring suites using Nagios
- Implement automated workflows for release process using C++ and Java programs and shell scripts

Relevant Skills

- Leadership and Mentorship: Multi-team management, cross-functional collaboration, mentoring junior engineers
- Cloud and Serverless Expertise: AWS services including DynamoDB, EC3, ECS, EKS, Elasticache, Kinesis, Lambda, Neptune, S3, SQS
- Security and Compliance: HIPAA, HITRUST, data governance, SAST and DAST testing
- Programming and Scripting: JavaScript, Node.js, Python, shell scripting
- Database Management: DynamoDB, Amazon Neptune, MySQL, PostgreSQL